STM32MP135-DK

MB1635

Table of contents

Sheet 1: Project overview (this page)

Sheet 2: Top

Sheet 3: STM32 MPU I/Os

Sheet 4: STM32 MPU power

Sheet 5: Board power

Sheet 6: DDR3L 16bits

Sheet 7: Ethernet1 - RMII - WoL support

Sheet 8: Ethernet2 - RMII

Sheet 9: WLAN & Bluetooth

Sheet 10: LCD 4.3"

Sheet 11: CSI camera bridge

Sheet 12: USB Host HUB

Sheet 13: USB Type-C DRP

Sheet 14: Expansion connector

Sheet 15: Peripherals

Sheet 16: IO expander

Sheet 17: ST-LINK V3E

Legend

General comment such as function title, configuration, ...

Text to be added to silkscreen.

Warning text.

Notes to generate the board layout.

Open Platform License Agreement

The Open Platform License Agreement (“Agreement”) is a binding legal contract between you ("You") and STMicroelectronics International N.V. (“ST”), a company incorporated under the laws of the Netherlands acting for the purpose of this Agreement through its Swiss branch 39, Chemin du Champ des Filles, 1228 Plan-les-Ouates, Geneva, Switzerland.

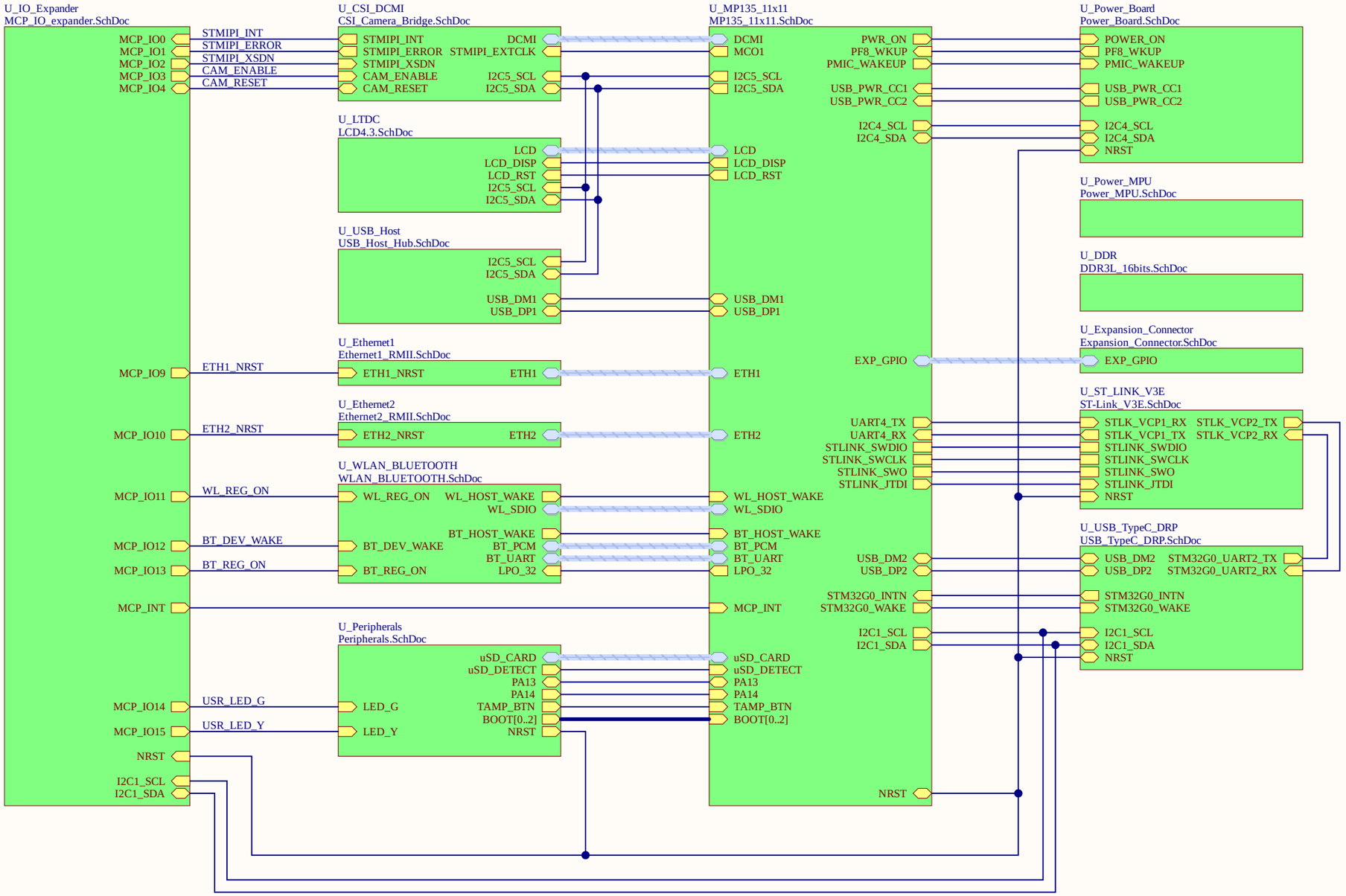
By using the enclosed reference designs, schematics, PC board layouts, and documentation, in hardcopy or CAD tool file format (collectively, the “Reference Material”), You are agreeing to be bound by the terms and conditions of this Agreement. Do not use the Reference Material until You have read and agreed to this Agreement terms and conditions. The use of the Reference Material automatically implies the acceptance of the Agreement terms and conditions.

The complete Open Platform License Agreement can be found on www.st.com/opla.

U_Top
MB1635-TOP.SchDoc

Title: **Project overview**			
Project: **STM32MP135-DK**			
Variant: MP135F			
Revision: E -02		Reference: MB1635	
Size: A4	Date: 16-JAN-2023	Sheet: 1 of 17	

TOP



Title: **TOP**		
Project: **STM32MP135-DK**		
Variant: MP135F		
Revision: E -02		Reference: MB1635
Size: A3	Date: 16-JAN-2023	Sheet: 2 of 17

STM32 I/Os

DUAL ETHERNET

ETH1: CLK, RX_CLK, MDINT, MDIO, MDC, TXD[0..1], TX_EN, RXD[0..1], CRS_DV
ETH1_CLK, ETH1_RX_CLK, ETH1_MDINT, ETH1_MDIO, ETH1_MDC, ETH1_TXD[0..1], ETH1_TX_EN, ETH1_RXD[0..1], ETH1_CRS_DV

ETH2: CLK, MDINT, MDIO, MDC, TXD[0..1], TX_EN, RXD[0..1], CRS_DV
ETH2_CLK, ETH2_MDINT, ETH2_MDIO, ETH2_MDC, ETH2_TXD[0..1], ETH2_TX_EN, ETH2_RXD[0..1], ETH2_CRS_DV

DISPLAY

LCD: LCD_R[2..7], LCD_G[2..7], LCD_B[2..7], LCD_HSYNC, LCD_VSYNC, LCD_CLK, LCD_DE, LCD_BL, LCD_INT

LCD_DISP, LCD_RST

WLAN-BLUETOOTH

BT_PCM: BT_PCM_CLK, BT_PCM_SDI, BT_PCM_SDO, BT_PCM_WS

BT_UART: USART2_TX, USART2_RX, USART2_CTS, USART2_RTS

WL_SDIO: SDMMC2_CK, SDMMC2_CMD, SDMMC2_D[0..3]

WL_HOST_WAKE, BT_HOST_WAKE, LPO_32

GPIO EXPANSION

EXP_GPIO: I2C1_SCL, I2C1_SDA, I2C5_SDA, I2C5_SCL, TIM3_CH4/DFSDM1_DATIN1, MCO2, TIM4_CH2, EXP_GPIO7, SPI5_NSS, SPI5_MISO, SPI5_MOSI, SPI5_SCK, TIM8_CH3, TIM14_CH1/UART8_RX, USART1_TX, USART1_RX, USART1_CTS, USART1_RTS, SAI1_SCKA, SAI1_FSA, SAI1_SDA, SAI1_SDB, DFSDM1_DATIN3, FDCAN2_TX, FDCAN2_RX, TAMP_IN8_OUT1, UART8_TX, DFSDM1_CKOUT

CAMERA

DCMI: DCMI_PIXCLK, DCMI_VSYNC, DCMI_HSYNC, DCMI_D[0..7]

MCO1 → STMIPI_EXTCLK

I2C

I2C1_SDA, I2C1_SCL — R63 2K4, R62 2K4 — VDD
I2C5_SDA, I2C5_SCL — R108 1K5, R82 1K5 — VDD

I2C PERIPHERAL	ADDRESS	I2C
IO expander	0100001X	I2C1
USB Type C	1010001X & 1010011X	I2C1
STPMIC	0110011X	I2C4
STMIPID02	0010100X	I2C5
ST camera module	0111100X	I2C5
USB HUB Type A	1010000X	I2C5
Touchscreen	1011101X	I2C5
Expansion connector	Shield dependant	I2C1&5

U14A

SAI1_SDB	Y3	PA0	PC0	W5	USART1_TX
ETH1_RX_CLK	AA3	PA1	PC1	Y9	ETH1_CRS_DV
ETH1_MDIO	Y4	PA2	PC2	AA10	USART1_RTS
USB_PWR_CC2	W9	PA3	PC3	W6	DFSDM1_CKOUT
SAI1_SCKA	AA6	PA4	PC4	Y7	ETH1_RXD0
SAI1_SDA	Y5	PA5	PC5	AA7	ETH1_RXD1
TAMP_BTN	U10	PA6	PC6	B14	LCD_HSYNC
USART1_CTS	V6	PA7	PC7	C13	LCD_G6
SPI5_MISO	P2	PA8	PC8	C18	SDMMC1_D0
DCMI_D0	A2	PA9	PC9	A20	SDMMC1_D1
STM32G0_WAKE	AA19	PA10	PC10	A18	SDMMC1_D2
ETH1_CLK	Y1	PA11	PC11	B18	SDMMC1_D3
ETH2_CRS_DV	F4	PA12	PC12	A19	SDMMC1_CK
EXTI13 PA13	T4	PA13	PC13	M3	PMIC_WAKEUP
EXTI14 PA14	W13	PA14	PC14-OSC32_IN	M2	
LCD_G7	B4	PA15	PC15-OSC32_OUT	L3	R103 0R
USART1_RX	W8	PB0	PD0	B1	DCMI_D1
TIM3_CH4	Y8	PB1	PD1	C5	I2C5_SCL
ETH2_MDIO	J3	PB2	PD2	C19	SDMMC1_CMD
SDMMC2_D2	C14	PB3	PD3	C2	DCMI_D5
SDMMC2_D3	C16	PB4	PD4	C7	USART2_RTS
BT_PCM_SDI	B13	PB5	PD5	C8	LCD_G4
LCD_B6	C1	PB6	PD6	D2	UART4_TX
DCMI_PIXCLK	C6	PB7	PD7	V1	MCO1 — TP54
DCMI_D6	D1	PB8	PD8	B5	UART4_RX — R72 10K — VDD
I2C4_SDA	A15	PB9	PD9	C9	LCD_CLK
BT_PCM_CLK	E13	PB10	PD10	D6	LCD_B2
ETH1_TX_EN	AA2	PB11	PD11	F3	DCMI_D4
LCD_R3	C11	PB12	PD12	A6	I2C1_SCL
BT_PCM_WS	D14	PB13	PD13	J2	R96 0R TIM4_CH2
SDMMC2_D0	B16	PB14	PD14	C4	LCD_R4
SDMMC2_D1	B15	PB15	PD15	E9	USART2_RX

STM32MP135FAF7

C93 6.8pF, X2 32.768KHz, C88 6.8pF

U14C

NRST	Y11	NRST	C118 100nF
PWR_ON	U17	PWR_ON	
PWR_LP	V19	PWR_LP	TP61
NJTRST	V11	PWR_NJTRST	TP63
BYPASS_REG1V8	W17		
USB_DP1	Y16	USB_D1_P	
USB_DM1	AA16	USB_D1_N	
USB_DP2	AA13	USB_D2_P	
USB_DM2	Y13	USB_D2_N	
USB_RREF	W12	R128 3K	

STM32MP135FAF7

U14B

LCD_B5	B6	PE0	PG0	A7	LCD_G5
UART8_TX	E8	PE1	PG1	R1	FDCAN2_TX
ETH2_RXD1	N1	PE2	PG2	V3	ETH1_MDC
SDMMC2_CK	B17	PE3	PG3	W1	FDCAN2_RX
DCMI_D3	H2	PE4	PG4	B10	LCD_VSYNC
TIM8_CH3	W10	PE5	PG5	G3	ETH2_MDC
TP50 MCO2	T2	PE6	PG6	C17	SDMMC2_CMD
LCD_R5	D8	PE7	PG7	A10	LCD_R2
I2C1_SDA	B9	PE8	PG8	H5	ETH2_CLK
LCD_R7	B7	PE9	PG9	F2	DCMI_VSYNC
	C10	PE10	PG10	G2	DCMI_D2
BT_HOST_WAKE R64 0R EXTI10 USART2_CTS	B2	PE11	PG11	R2	ETH2_TXD1
LCD_BL	A4	PE12	PG12	W2	MCP_INT EXTI12
LCD_R6	A3	PE13	PG13	AA9	ETH1_TXD0
DCMI_D7	C3	PE14	PG14	Y10	ETH1_TXD1
I2C4_SCL	A9	PE15	PG15	H4	ETH2_MDINT EXTI15
EXTI0 WL_HOST_WAKE	A16	PF0	PH0-OSC_IN	T3	HSE_IN
LCD_B7	B11	PF1	PH1-OSC_OUT	U2	HSE_OUT
LCD_B3	D11	PF2	PH2	E2	LCD_RST
LCD_G3	A13	PF3	PH3	R3	SPI5_MOSI
ETH2_RXD0	N2	PF4	PH4	Y19	PH4 EXTI4
EXTI5 LCD_INT	B3	PF5	PH5	W18	STLINK_SWO
ETH2_TX_EN	G1	PF6	PH6	N3	I2C5_SDA
ETH2_TXD0	P3	PF7	PH7	J1	SPI5_SCK
PF8_WKUP	F1	PF8	PH8	E3	DCMI_HSYNC
TIM14_CH1 R85 0R	J5	PF9	PH9	C12	LCD_DE
EXP_GPIO7	H3	PF10	PH10	E14	BT_PCM_SDO
SAI1_FSA	Y6	PF11	PH11	K3	SPI5_NSS
USB_PWR_CC1	AA4	PF12	PH12	D3	USART2_TX
DFSDM1_DATIN3	W4	PF13	PH13	A12	LCD_G2
STLINK_SWCLK	Y2	PF14	PH14	B8	LCD_B4
STLINK_SWDIO	Y12	PF15		P5	TAMP_IN8_OUT1
			PI0	P4	LPO_32
VDD — TP57	AA12		PI1	N5	STM32G0_INTN EXTI2
PDR_ON	W3		PI2	L4	ETH1_MDINT EXTI3
PWR_CPU_ON			PI3	Y17	BOOT0
			PI4-BOOT0	AA15	BOOT1
			PI5-BOOT1	Y15	BOOT2
			PI6-BOOT2	AA20	LCD_DISP
			PI7		

STM32MP135FAF7

C98 6.8pF, X4 24MHz, C103 6.8pF — HSE_IN, HSE_OUT

BOOT MODE

BOOT[0..2]

POWER - PMIC

USB_PWR_CC1, USB_PWR_CC2, PWR_ON, NRST, PMIC_WAKEUP, PF8_WKUP

I2C4_SDA, I2C4_SCL — R71 1K5, R75 1K5 — VDD

LED & BUTTON

PA13, PA14, TAMP_BTN

USB Host & Type C

USB_DP1, USB_DM1 — USB_D1_P, USB_D1_N
USB_DP2, USB_DM2, STM32G0_WAKE, STM32G0_INTN — USB_D2_P, USB_D2_N, STM32G0_WAKE, STM32G0_INTN

STLINK

UART4_TX, UART4_RX
STLINK_SWDIO, STLINK_SWCLK, STLINK_SWO

IO EXPANDER

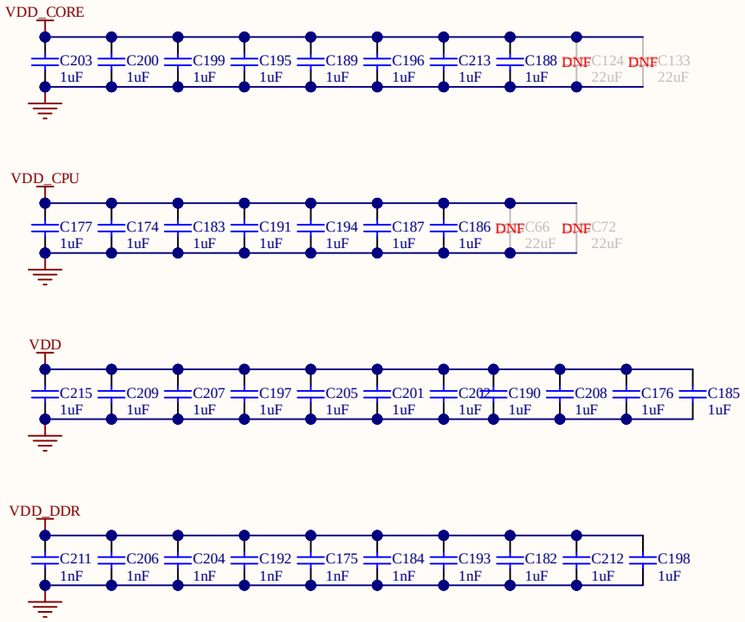
MCP_INT

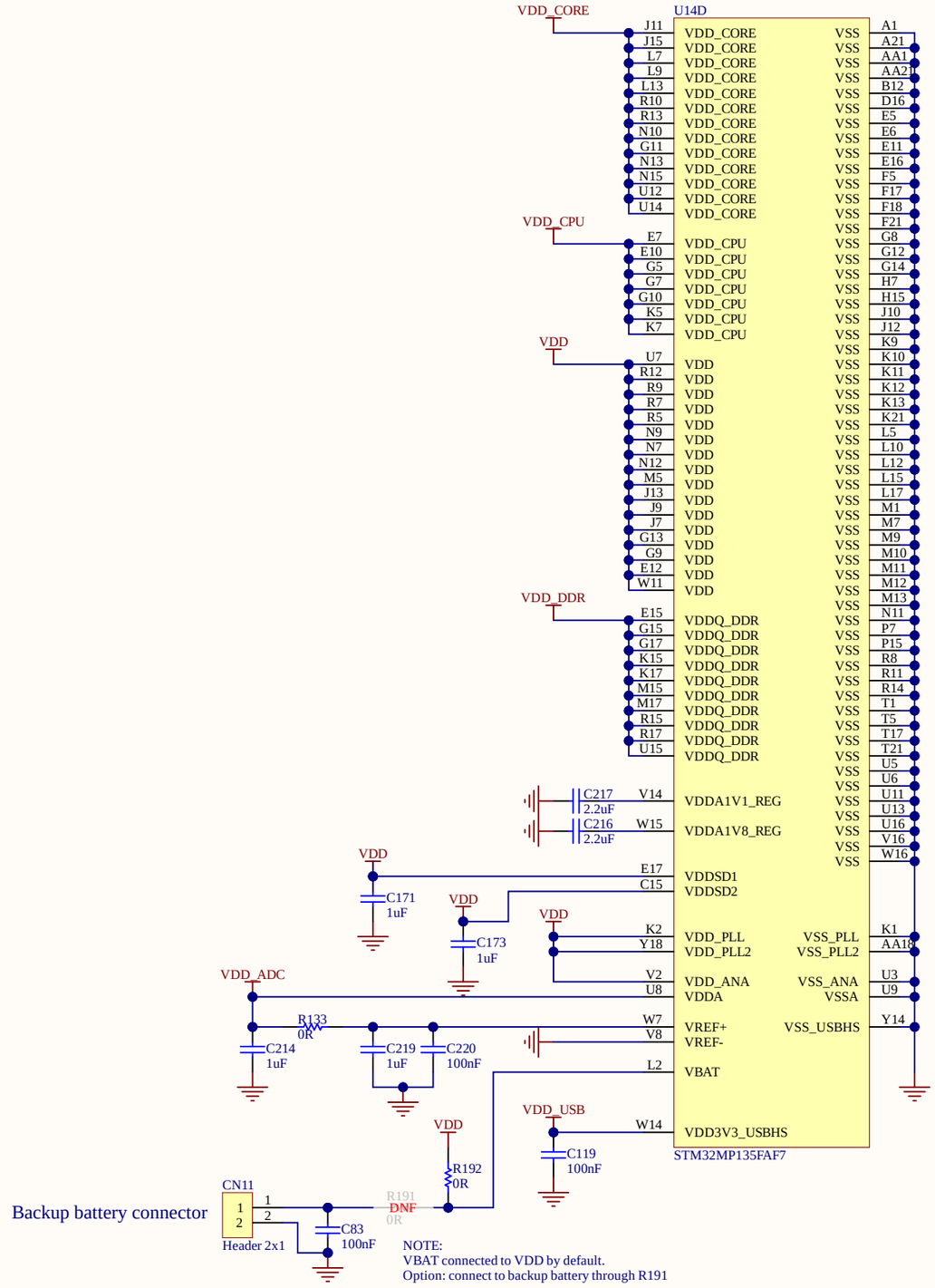
uSD CARD

uSD_CARD: SDMMC1_D[0..3], SDMMC1_CK, SDMMC1_CMD
uSD_DETECT, STLINK_JTDI — R188 0R PH4, R189 0R DNF

Title:	**STM32 MPU I/Os**
Project:	**STM32MP135-DK**
Variant:	MP135F
Revision:	E -02
Reference:	MB1635
Size: A3	Date: 16-JAN-2023
Sheet: 3	of 17

STM32 Power

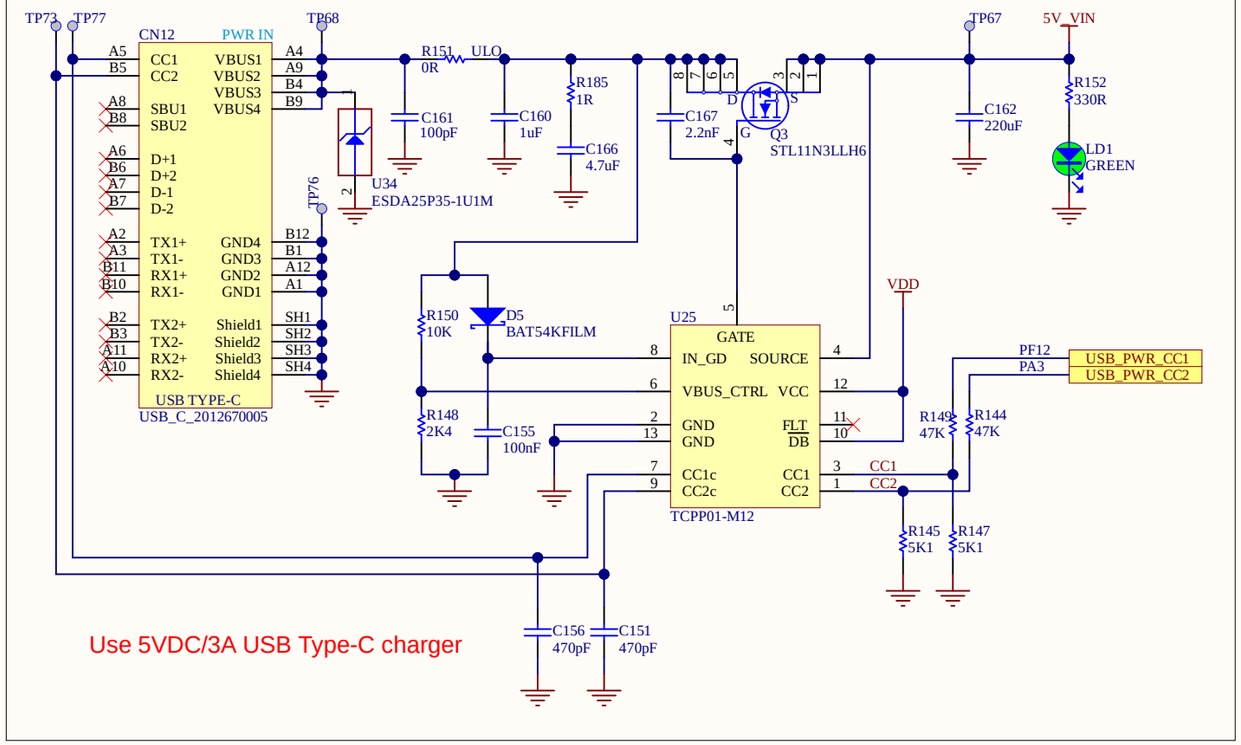




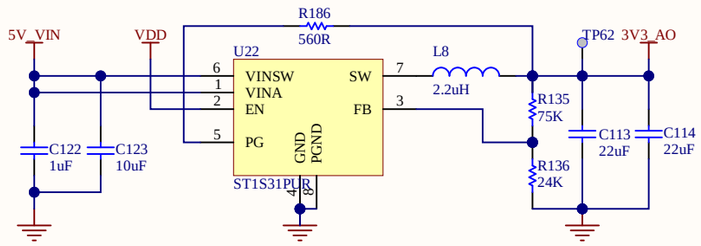
Title: **STM32 MPU power**		
Project: **STM32MP135-DK**		
Variant: MP135F		
Revision: E-02		Reference: MB1635
Size: A3	Date: 16-JAN-2023	Sheet: 4 of 17

Board Power

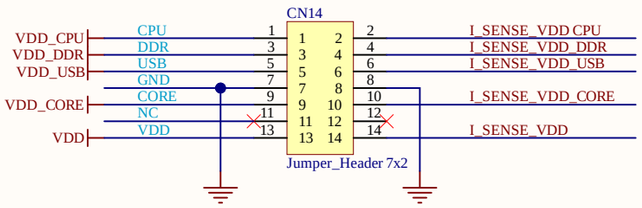
BOARD SUPPLY



ALWAYS-ON 3V3

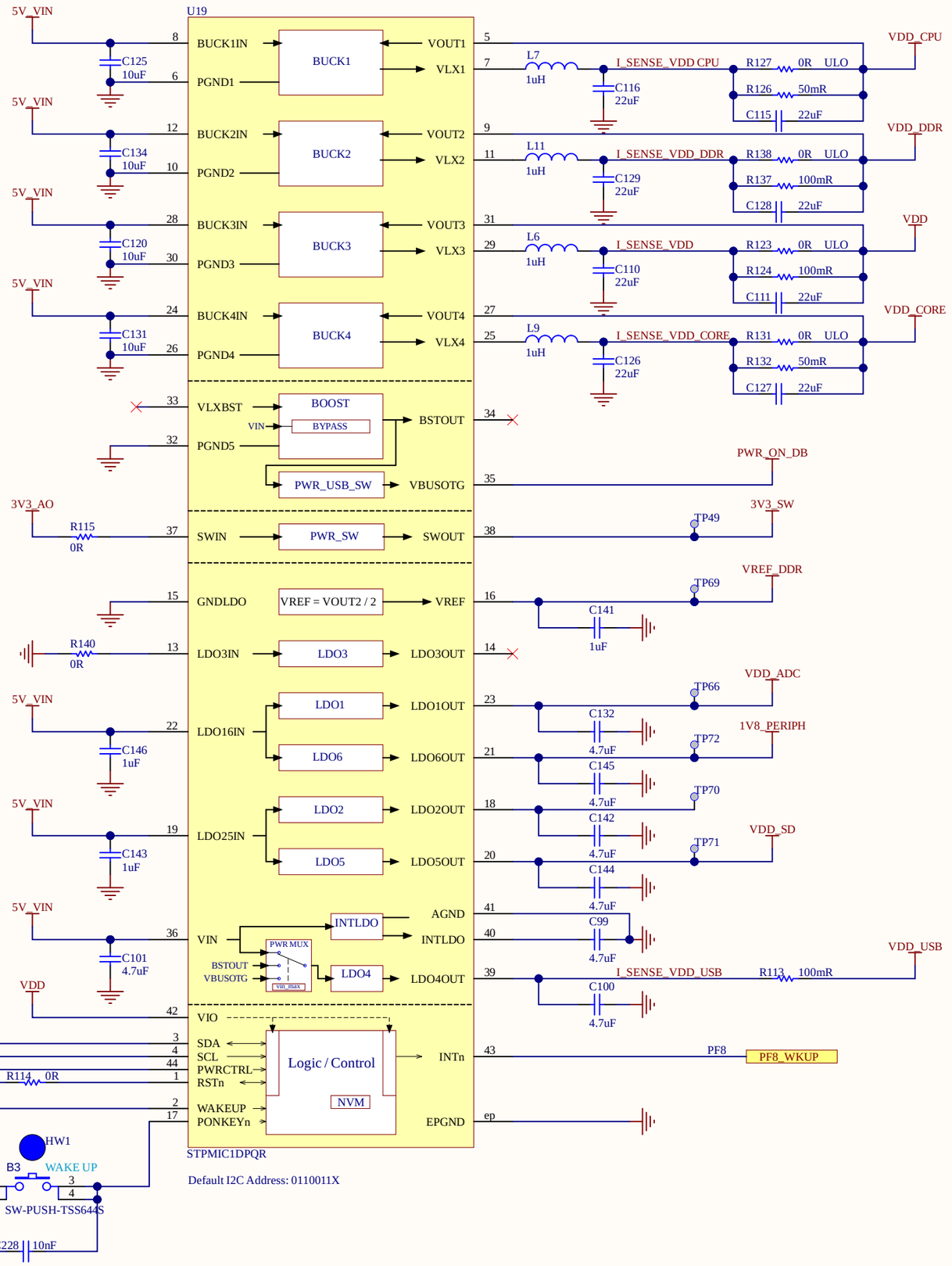


POWER MEASUREMENT



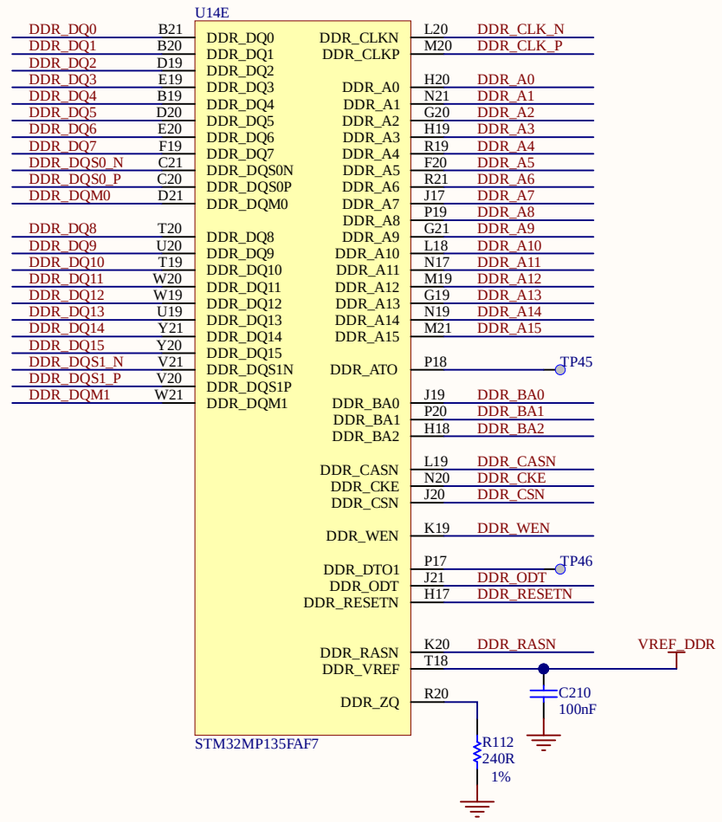
NOTE:
- Only ULO resistors from PMIC buck outputs have to be removed to get the image of current through 100mOhms resistors via voltage measurements directly at the corresponding header terminals.
- 22uF capacitors must remain to ensure low impedance path at high frequencies.

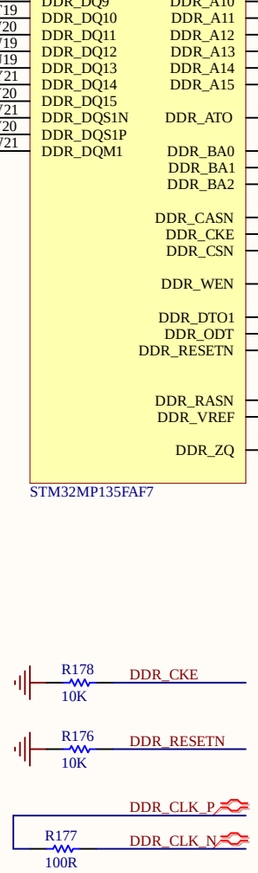
Example for VDD_CORE: remove R131 and measure voltage between pins 9 & 10 of header terminals above.

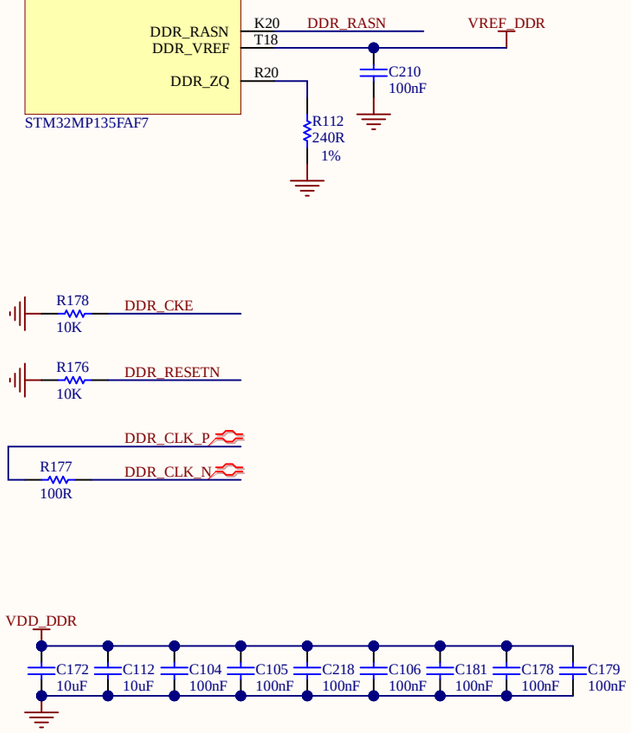


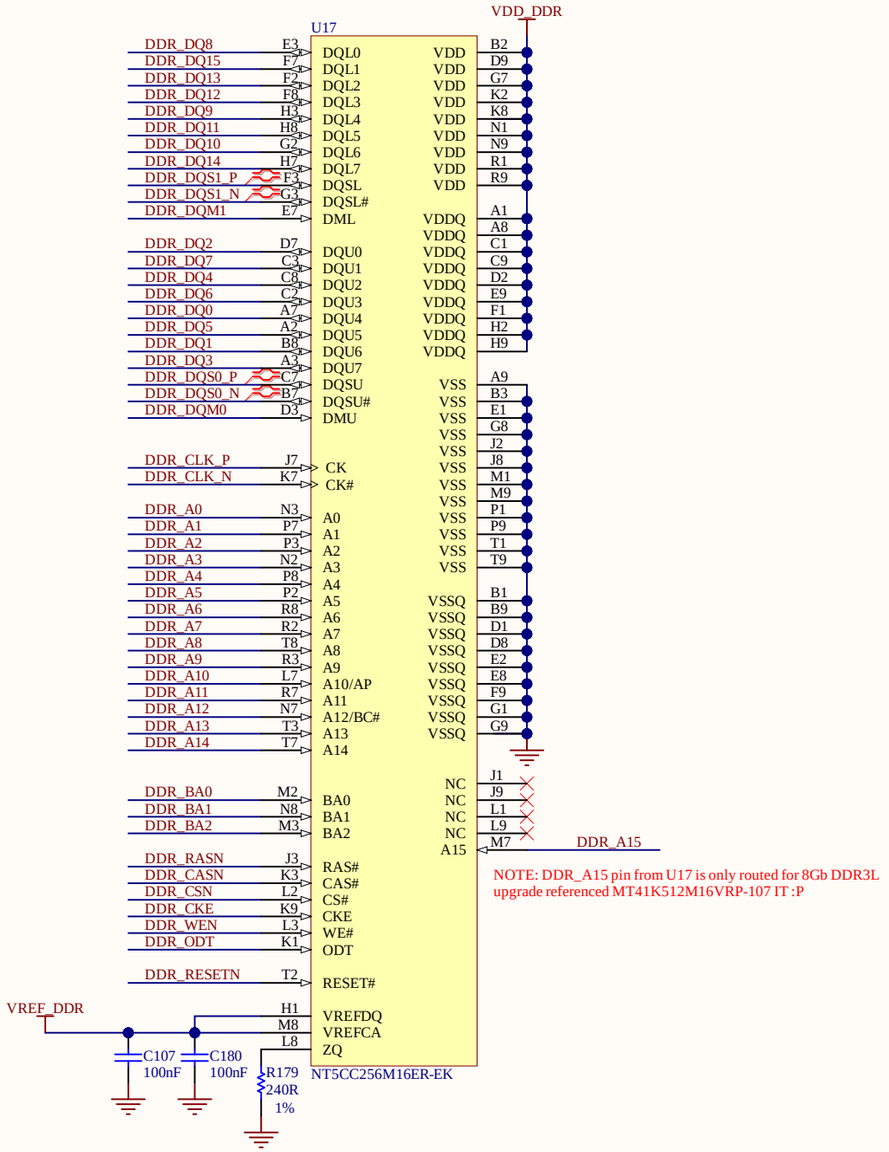
Title: **Board power**		
Project: **STM32MP135-DK**		
Variant: MP135F		
Revision: E -02		Reference: MB1635
Size: A3	Date: 16-JAN-2023	Sheet: 5 of 17

DDR3L 16bits

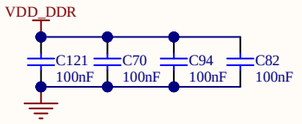






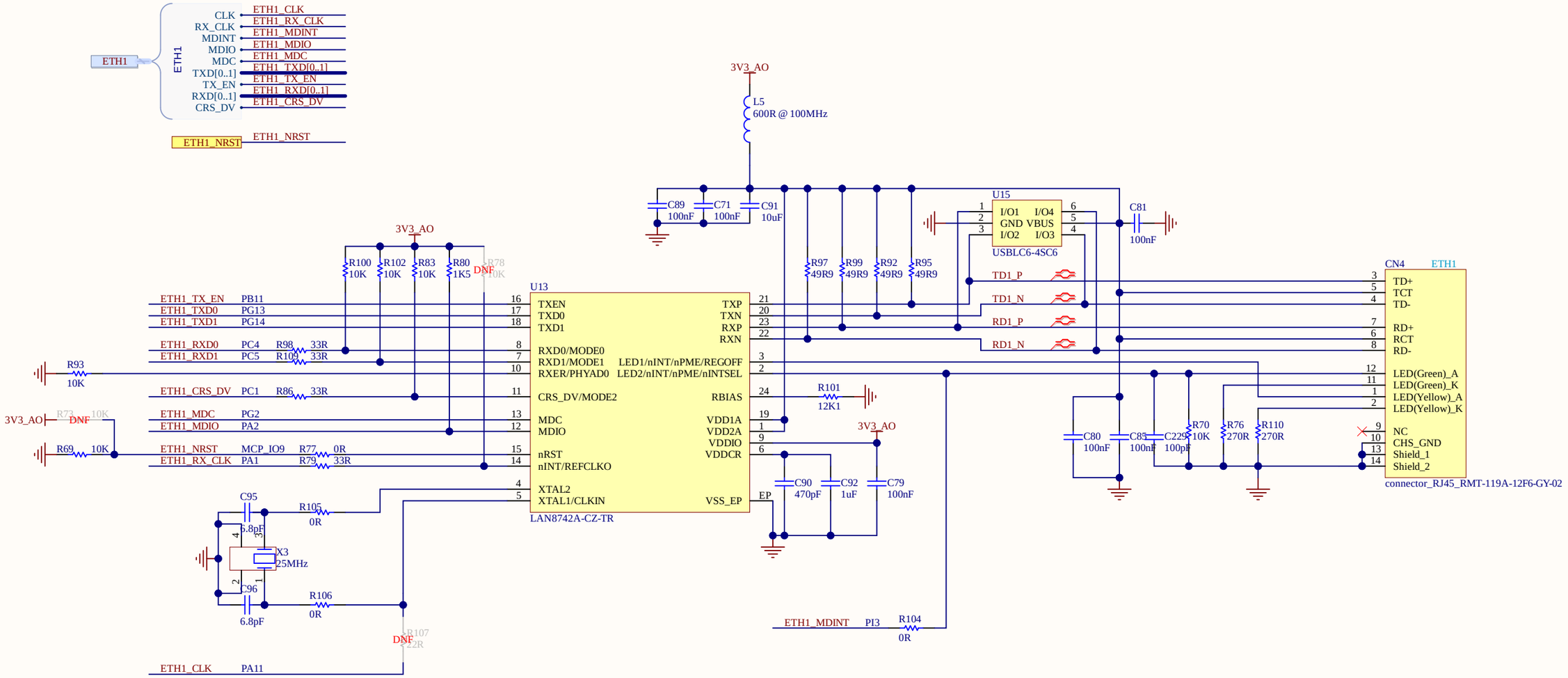


To connect to VDD_DDR power plane



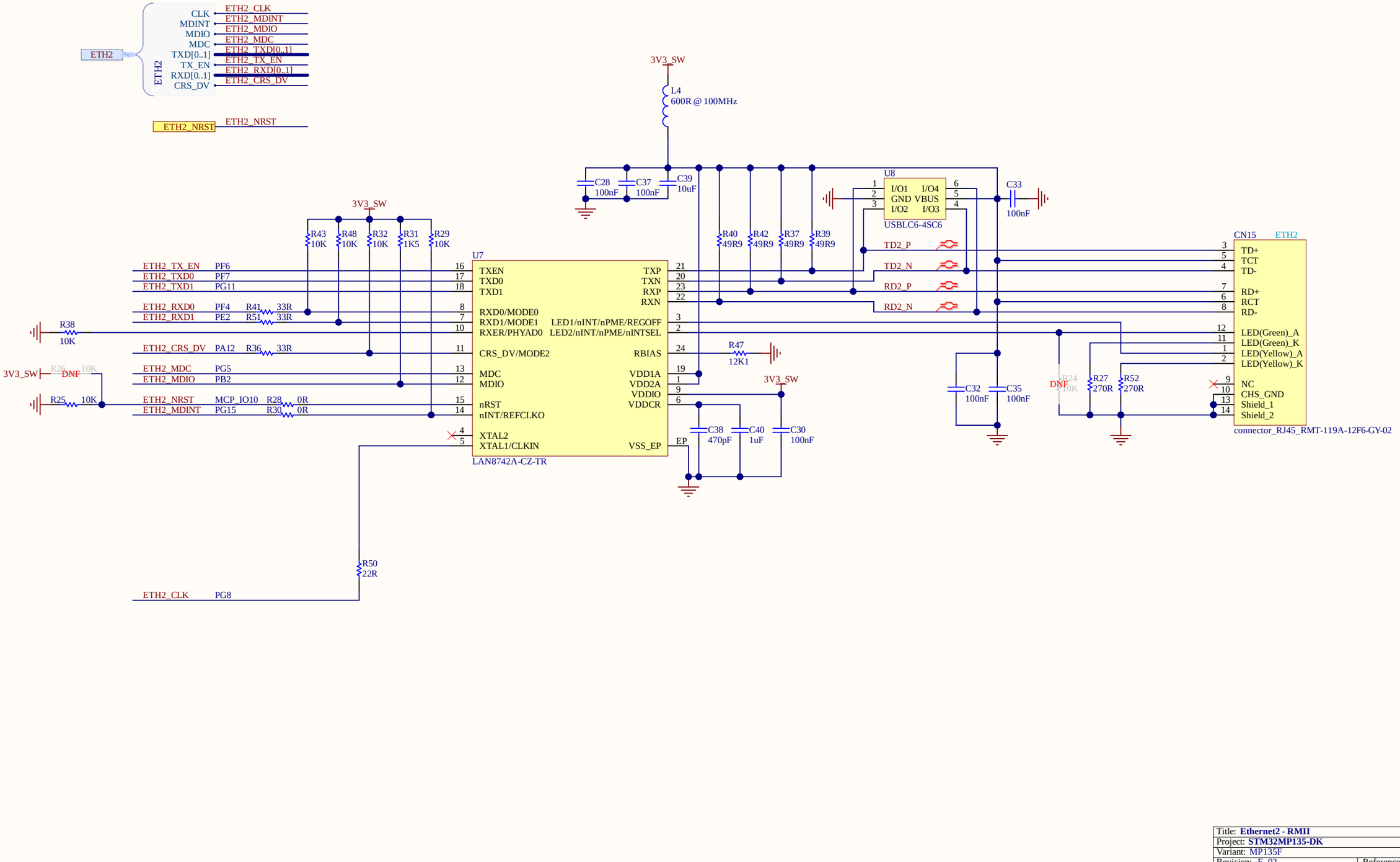
Title: **DDR3L 16bits**		
Project: **STM32MP135-DK**		
Variant: MP135F		
Revision: E-02		Reference: MB1635
Size: A3	Date: 16-JAN-2023	Sheet: 6 of 17

ETHERNET1 - RMII - WoL support



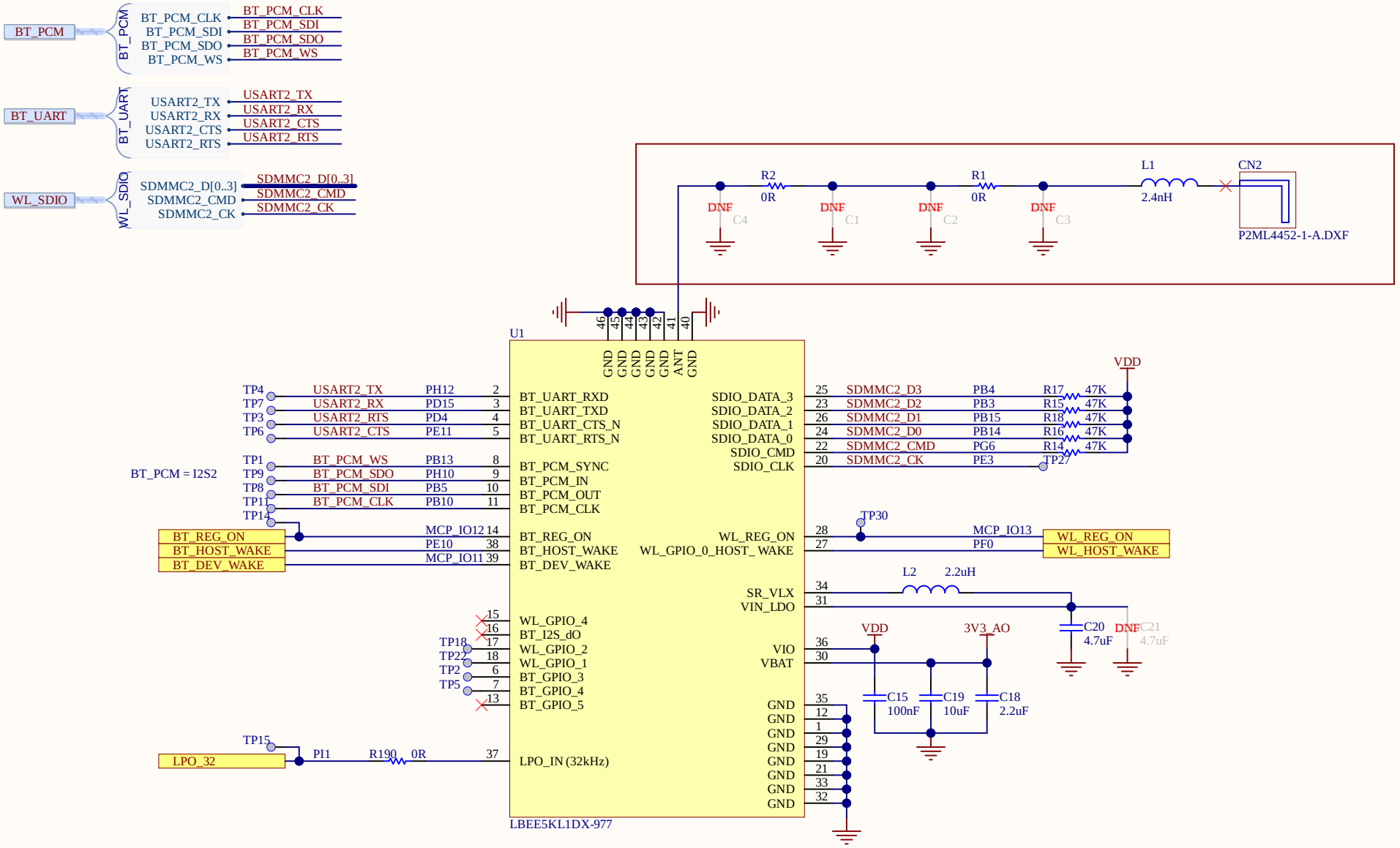
Title: **Ethernet1 - RMII - WoL support**			
Project: **STM32MP135-DK**			
Variant: MP135F			
Revision: E-02		Reference: MB1635	
Size: A3	Date: 16-JAN-2023	Sheet: 7 of 17	

ETHERNET2 - RMII



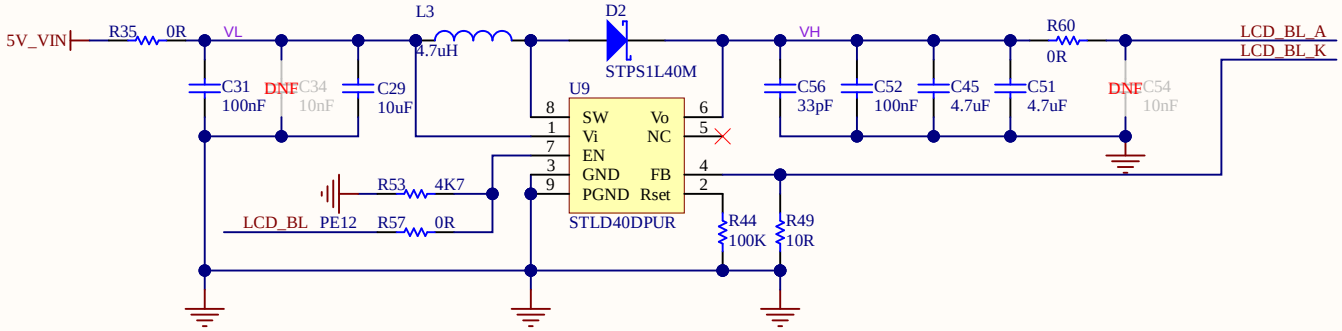
Title: **Ethernet2 - RMII**			
Project: **STM32MP135-DK**			
Variant: MP135F			
Revision: E -02		Reference: MB1635	
Size: A3	Date: 16-JAN-2023	Sheet: 8 of 17	

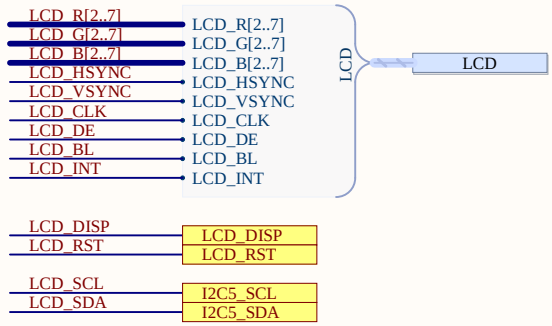
WLAN & Bluetooth

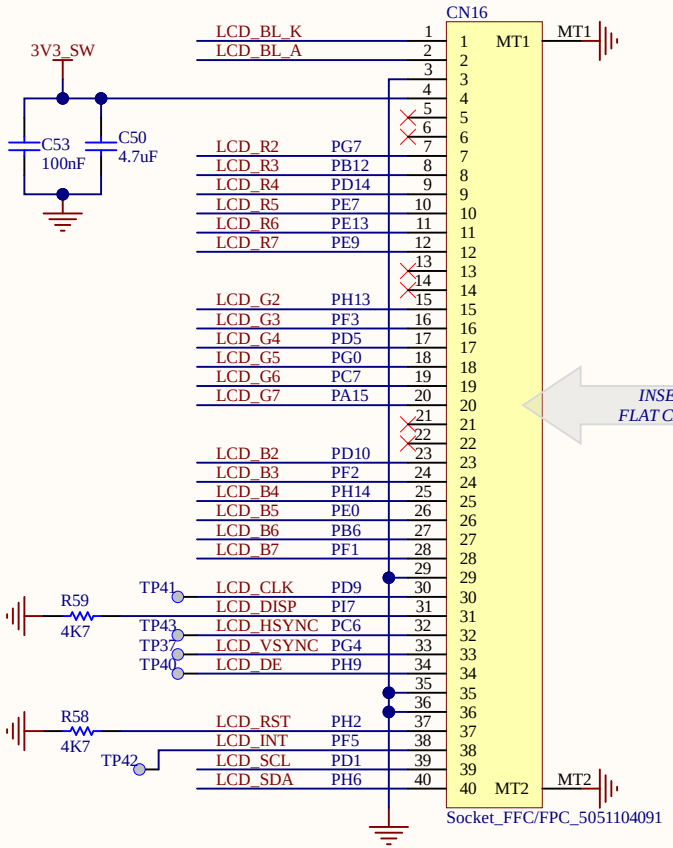


Title: **WLAN & Bluetooth**		
Project: **STM32MP135-DK**		
Variant: MP135F		
Revision: E-02		Reference: MB1635
Size: A4	Date: 16-JAN-2023	Sheet: 9 of 17

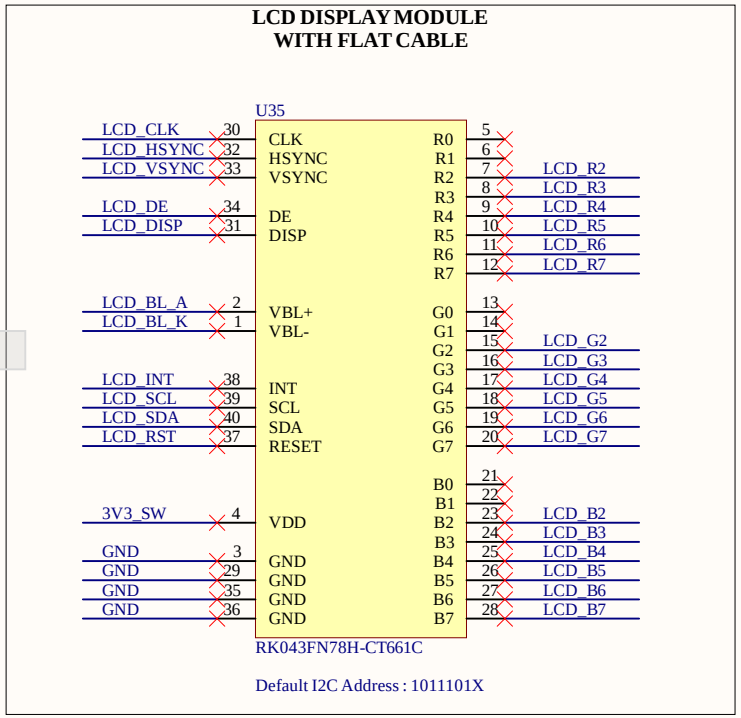
LCD 4.3"





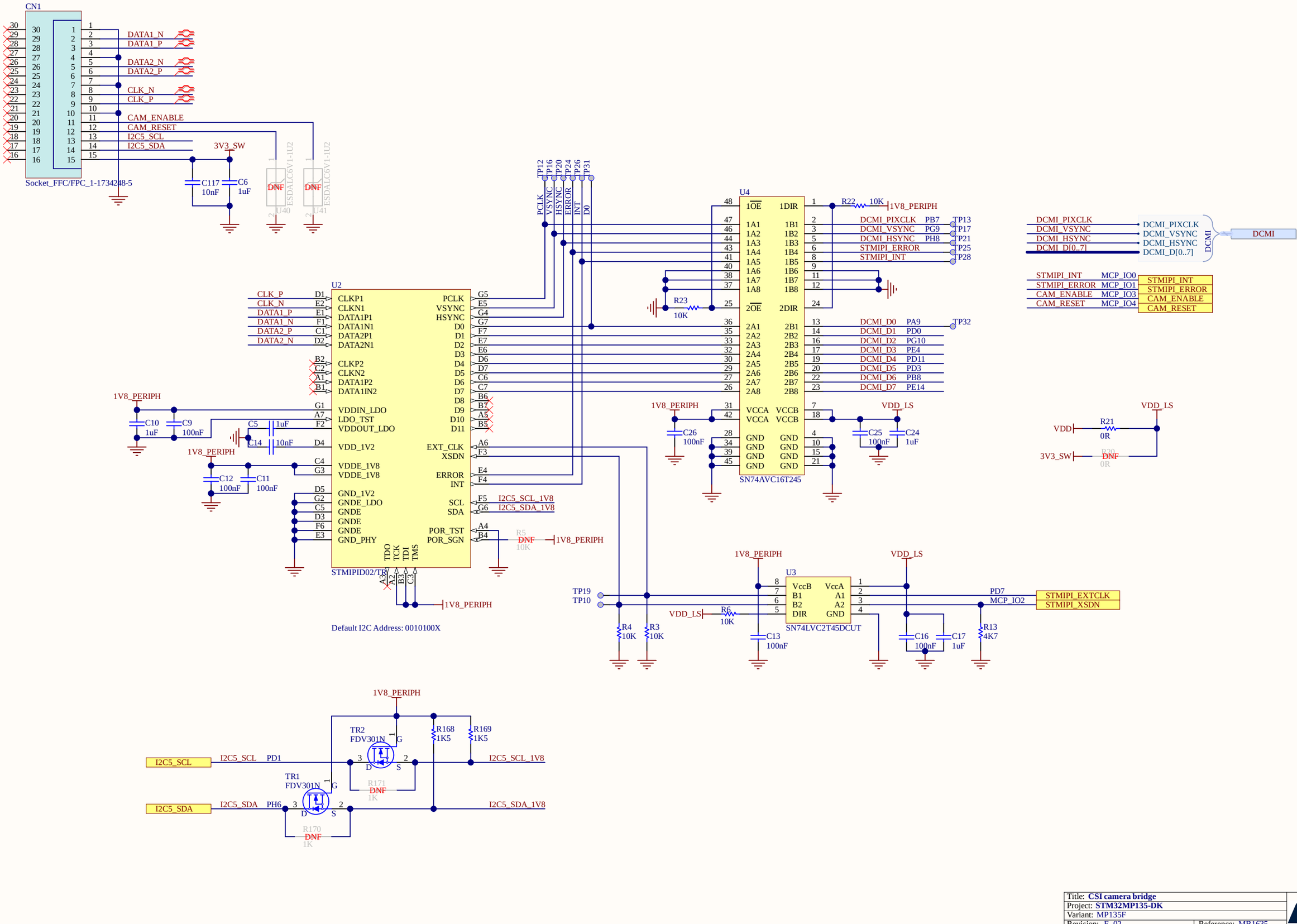


INSERT FLAT CABLE



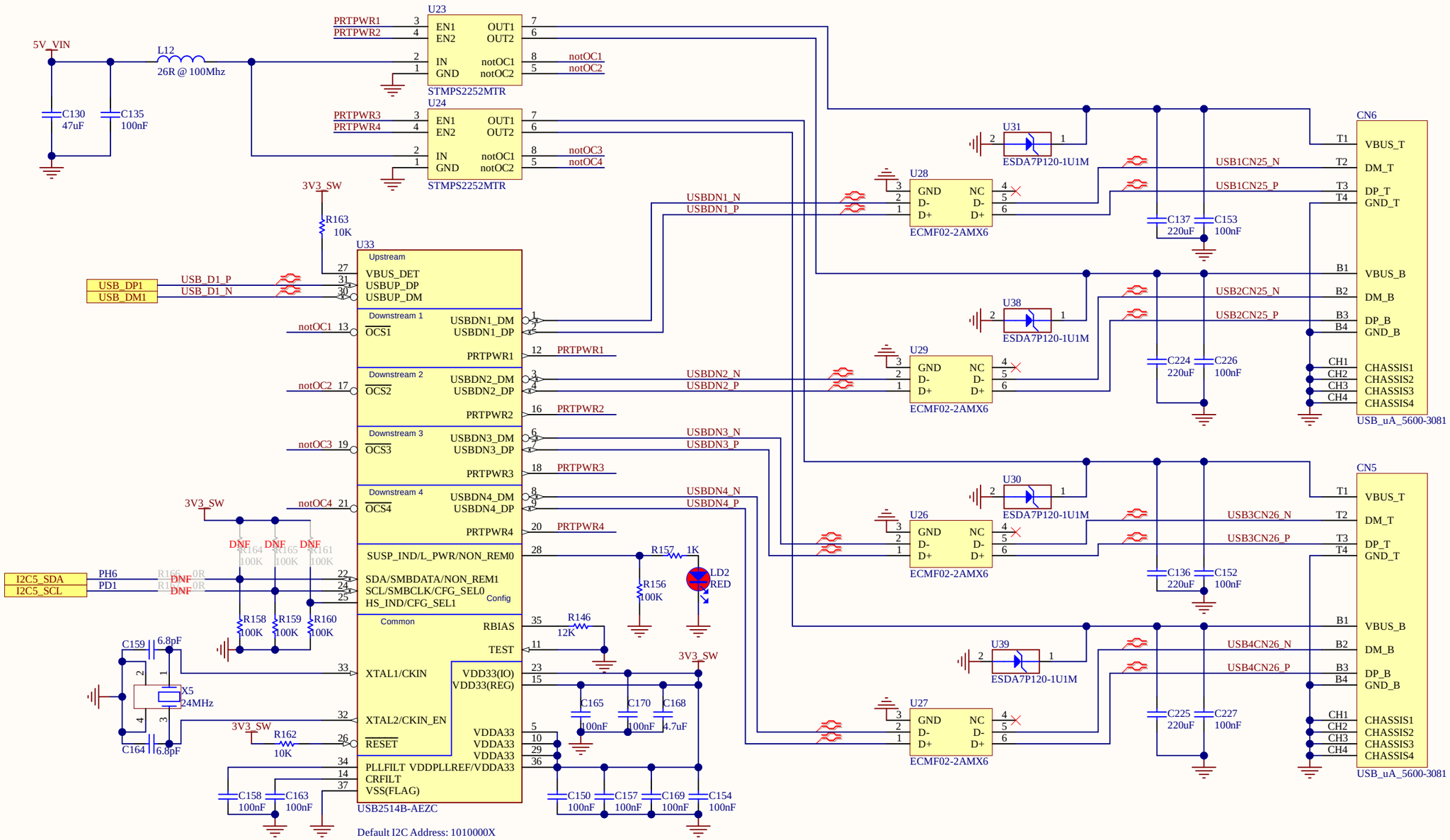
Title: **LCD 4.3"**		
Project: **STM32MP135-DK**		
Variant: MP135F		
Revision: E-02		Reference: MB1635
Size: A4	Date: 16-JAN-2023	Sheet: 10 of 17

CSI CAMERA



Title: **CSI camera bridge**		
Project: **STM32MP135-DK**		
Variant: MP135F		
Revision: E -02		Reference: MB1635
Size: A3	Date: 16-JAN-2023	Sheet: 11 of 17

USB HOST HUB



Title: **USB Host HUB**		
Project: **STM32MP135-DK**		
Variant: MP135F		
Revision: E-02		Reference: MB1635
Size: A3	Date: 16-JAN-2023	Sheet: 12 of 17

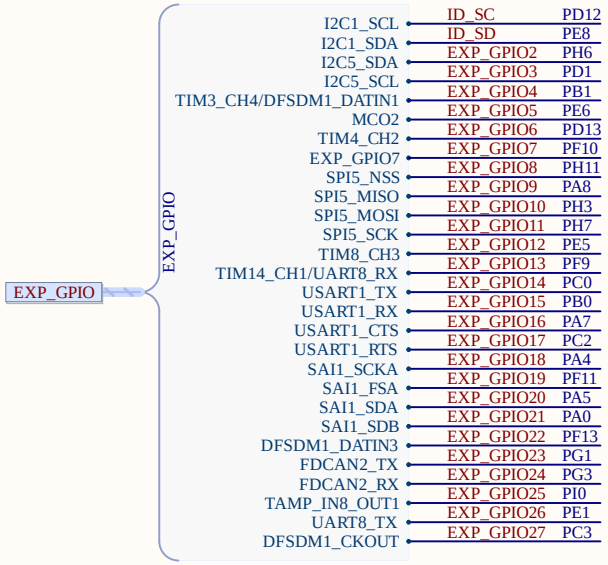
USB Type-C DRP

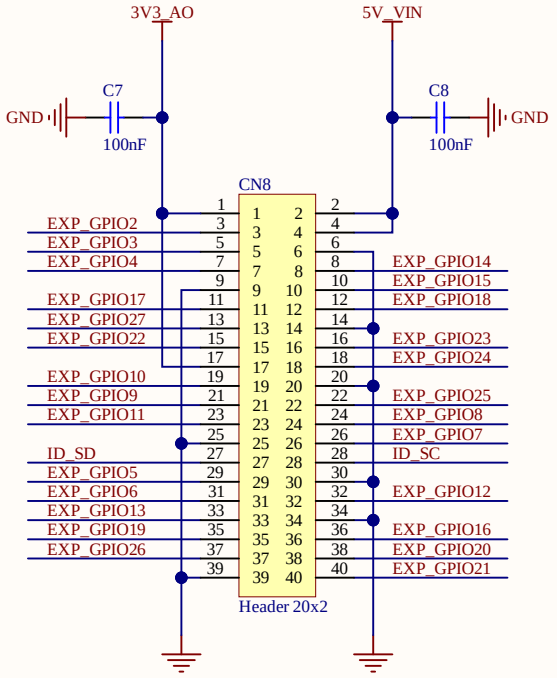
Source/Sink

5V_VIN
SOURCE
SRC_GDCS
C86 value is set due to significant 5V_VIN decoupling capacitors.
C86 1nF
SRC_GDCG
Q1
D1 S1 S2 D2 G1 G2
STL40DN3LLH5
Note:
R111 resistor value considers copper track resistance to get a 1.9A overcurrent protection.
R111 12mR
PWR_ON_DB
SINK
SNK_GDCS
C102 100nF
Q2
D1 S1 S2 D2 G1 G2
STL40DN3LLH5
DNF C87 22pF
SNK_GDCG
R142 33K
R143 8K2
VBUS_MEASURE
R129 13K
R134 2K
VDD
DNF R141 10K
D4
NRST
BAT60JFILM
C140 10nF
C139 4.7uF
C138 100nF
R187 10K
U21
VDD/VDDA
PF2-NRST
PC14-OSC32_IN
PC15-OSC32_OUT
PC6
PD0
PD1
PD2
PD3
VSS/VSSA
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA11 [PA9]
PA12 [PA10]
PA13
PA14-BOOT0
PB0
PB6
PB7
PB8
PB15
STM32G071G8U6N
STM32G0_UART2_TX
STM32G0_UART2_RX
VBUS_MEASURE
STM32G0_I2C2_SCL R121 1K5
STM32G0_I2C2_SDA R120 1K5
STM32G0_SWDIO
STM32G0_SWCLK
ISENSE_MEASURE
I2C1_SCL
I2C1_SDA
STM32G0_INTN
TP51 USBC_INTn
TP56 USBC_EN
STM32G0_WAKE
PA10
TP58
TP59
Default I2C address
Firmware update | 1010001X
UCSI | 1010011X
VDD
C109 4.7uF
C108 100nF
U20
VCC_VCONN
EN
IANA
CC1
CC2
FLGn
SDA
SCL
GND
GDCS GDCG GDPS GDPG
ISENSE
VBUSc
VSENSE
CC1c
CC2c
CBIAS
I2C_ADD
USBC_EN
ISENSE_MEASURE
USBC_INTn
STM32G0_I2C2_SDA
STM32G0_I2C2_SCL
TP53
TP52
TP55
C149 100nF
C148 470pF
C147 470pF
TCPP03-M20
Default I2C Address: 0110100X
TP65
TP64
CN7
CC1 CC2 SBU1 SBU2 D+1 D+2 D-1 D-2
TX1+ TX1- RX1+ RX1-
TX2+ TX2- RX2+ RX2-
VBUS1 VBUS2 VBUS3 VBUS4
GND4 GND3 GND2 GND1
Shield1 Shield2 Shield3 Shield4
USB TYPE-C
USB_C_2012670005
TP74
TP75
U36
ESDA25P35-1U1M
C221 100pF
C222 10uF
CN13
STM32G0_SWDIO
STM32G0_SWCLK
VDD
DNF
Header 4x1
U32
USB_DP2
USB_DM2
USB_D2_P
USB_D2_N
D+ D- GND NC
ECMF02-2AMX6
USB_DCONN_P
USB_DCONN_N

Title: **USB Type-C DRP**			
Project: **STM32MP135-DK**			
Variant: MP135F			
Revision: E -02		Reference: MB1635	
Size: A3	Date: 16-JAN-2023	Sheet: 13 of 17	

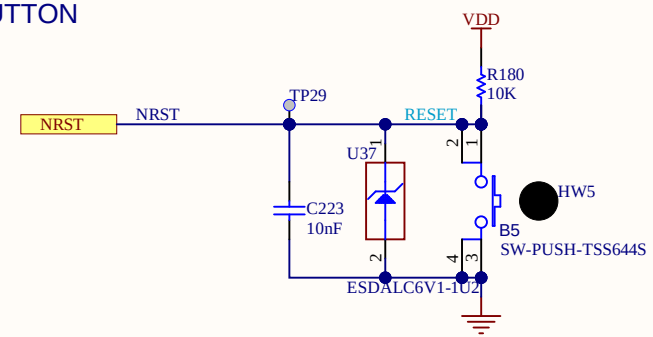
EXPANSION CONNECTOR





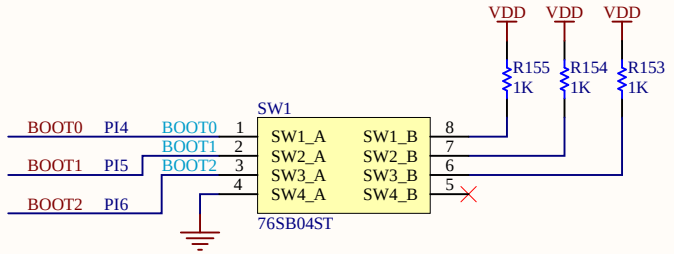
Title: **Expansion connector**		
Project: **STM32MP135-DK**		
Variant: MP135F		
Revision: E-02		Reference: MB1635
Size: A4	Date: 16-JAN-2023	Sheet: 14 of 17

RESET BUTTON

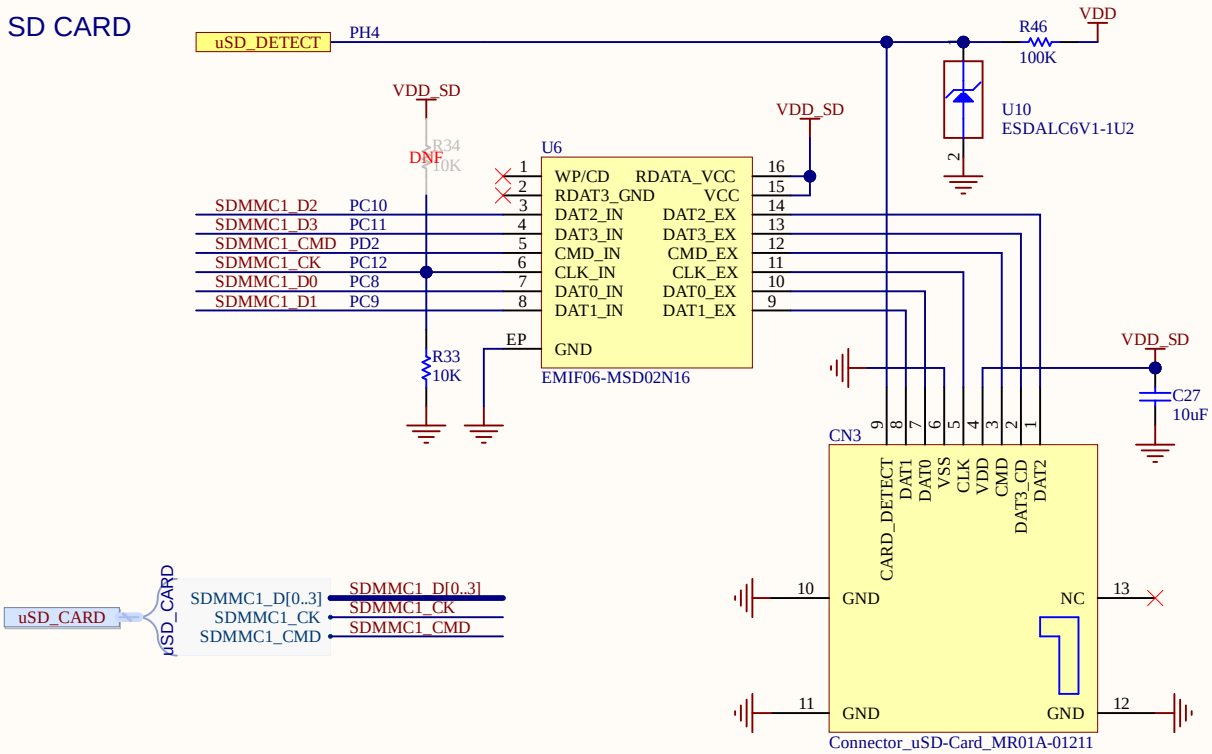


BOOT MODE

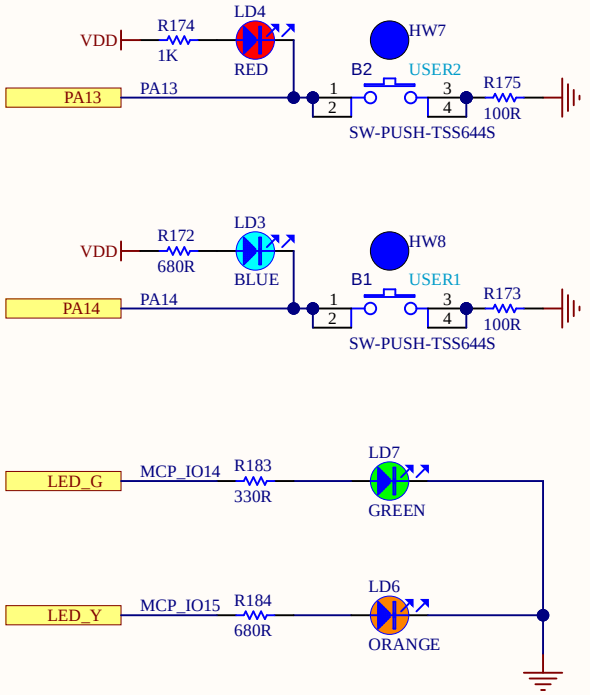
BOOT MODE	BOOT2	BOOT1	BOOT0
SD-Card	1	0	1
Forced UART/USB	0	0	0
Reserved	1	0	0



MICRO SD CARD



LED & BUTTONS

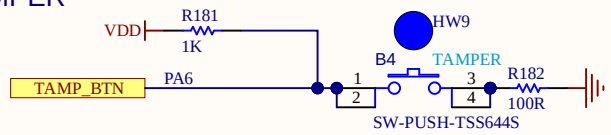


HW3
STICKER PRODUCT
Sticker product

HW2
STICKER BOARD
Sticker board

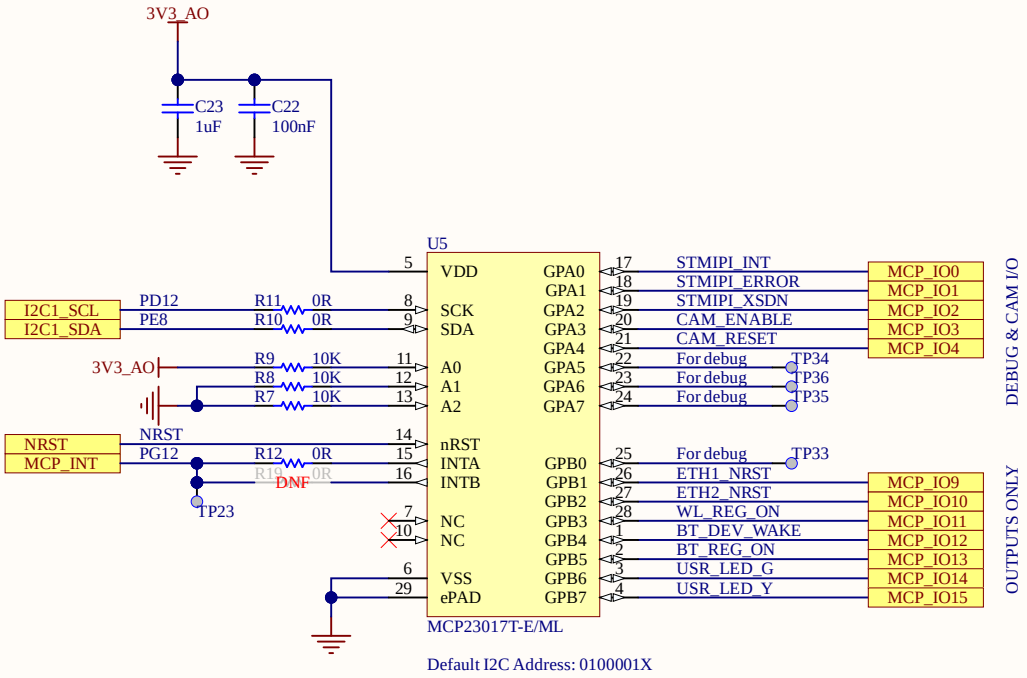
HW4
PCB
MBxxxxy

TAMPER



Title: **Peripherals**			
Project: **STM32MP135-DK**			
Variant: MP135F			
Revision: E-02		Reference: MB1635	
Size: A4	Date: 16-JAN-2023	Sheet: 15 of 17	

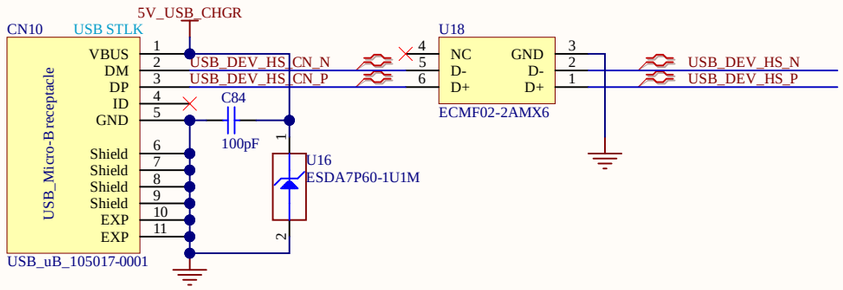
IO expander



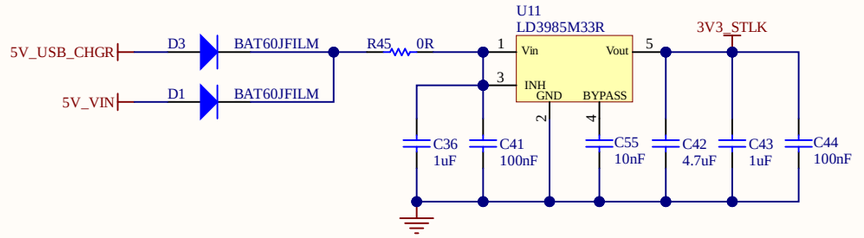
Title: **IO expander**		
Project: **STM32MP135-DK**		
Variant: MP135F		
Revision: E-02		Reference: MB1635
Size: A4	Date: 16-JAN-2023	Sheet: 16 of 17

ST-LINKV3E

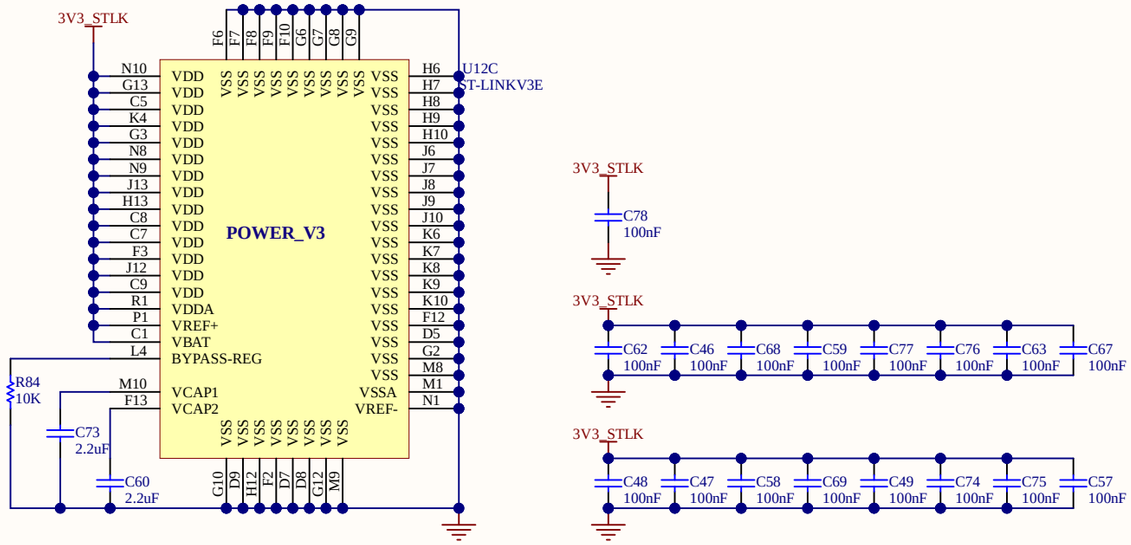
ST-LINKV3E USB HS & 5V USB CHARGER



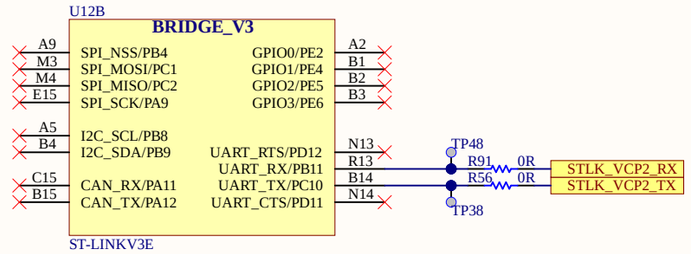
3V3 for ST-LINKV3E Only



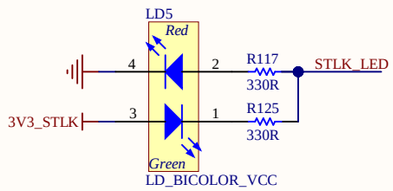
ST-LINKV3E MCU POWER



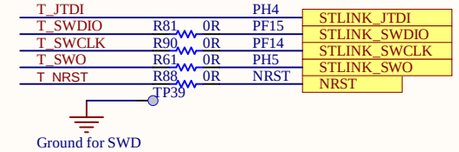
ST-LINKV3E Bridges (SPI/I2C/CAN/USART/GPIOs)



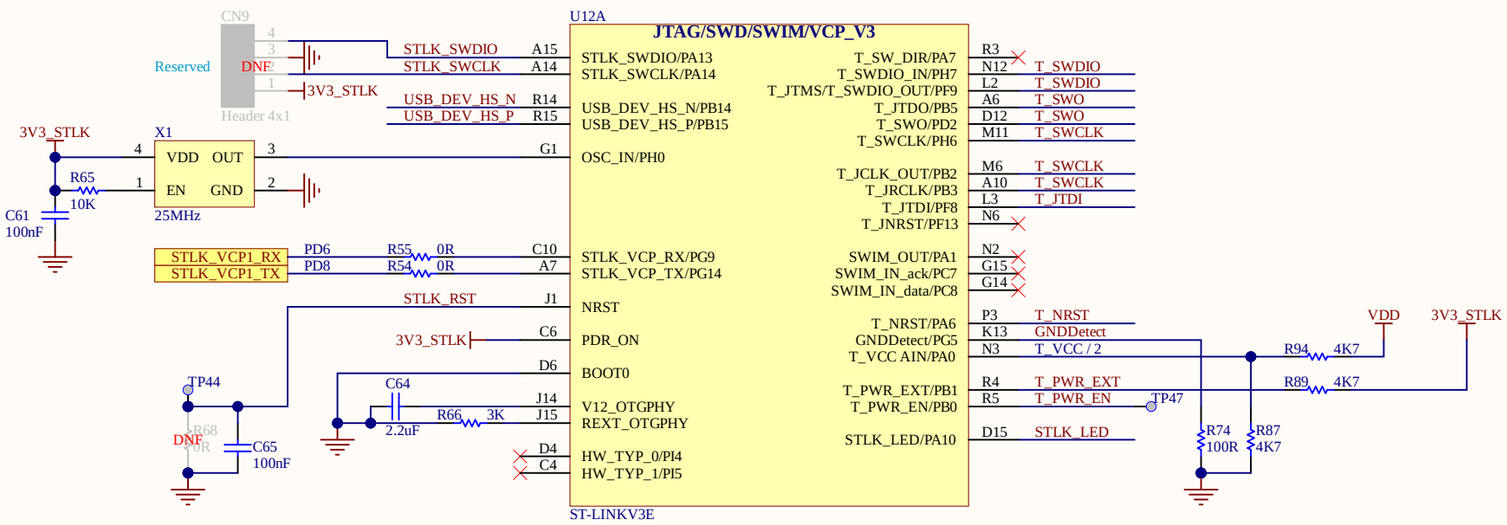
ST-LINKV3E Communication LEDs



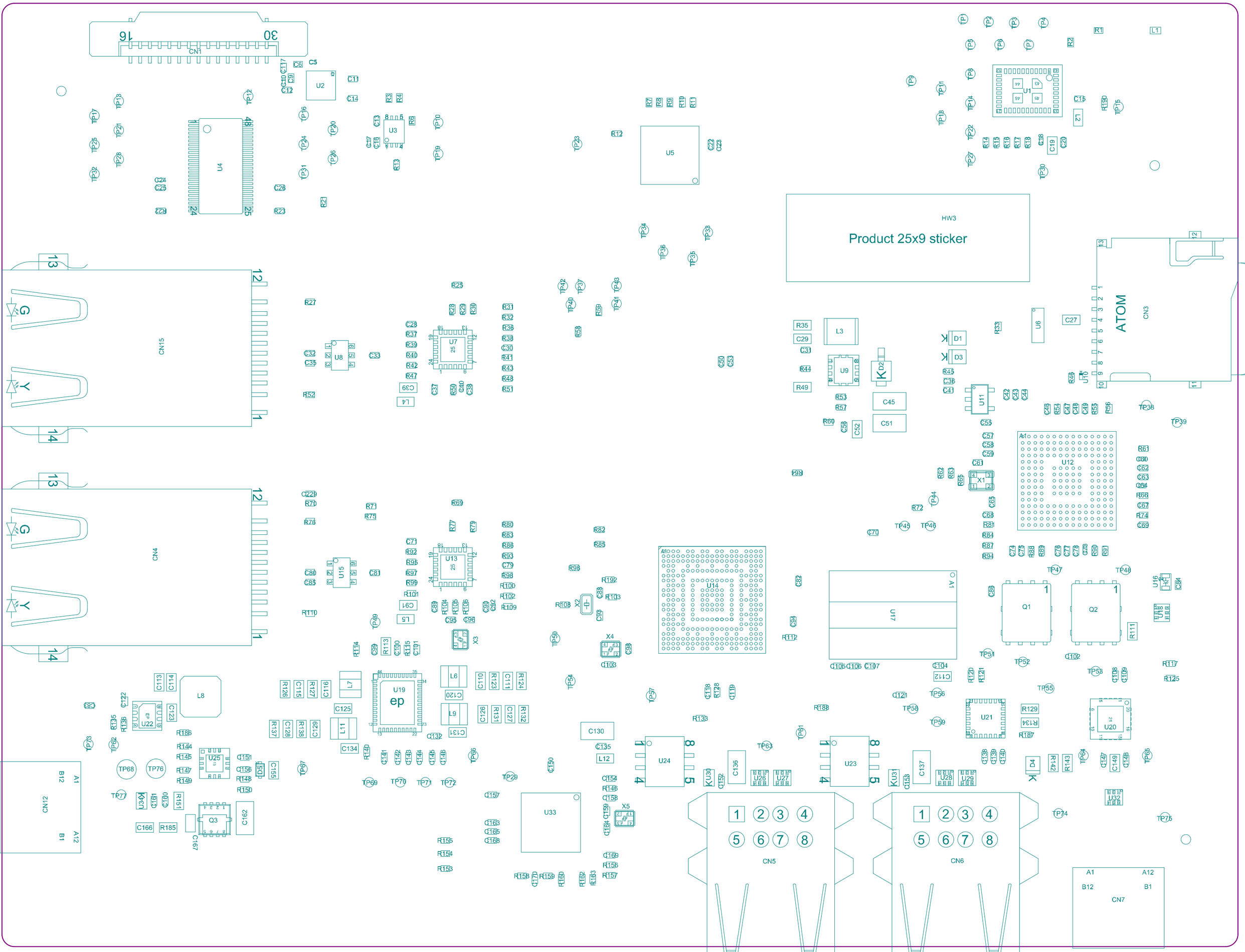
SWD INTERACE



ST-LINKV3E JTAG/SWD/SWIM Debug Module



Title: **ST-LINK V3E**		
Project: **STM32MP135-DK**		
Variant: MP135F		
Revision: E-02		Reference: MB1635
Size: A3	Date: 16-JAN-2023	Sheet: 17 of 17



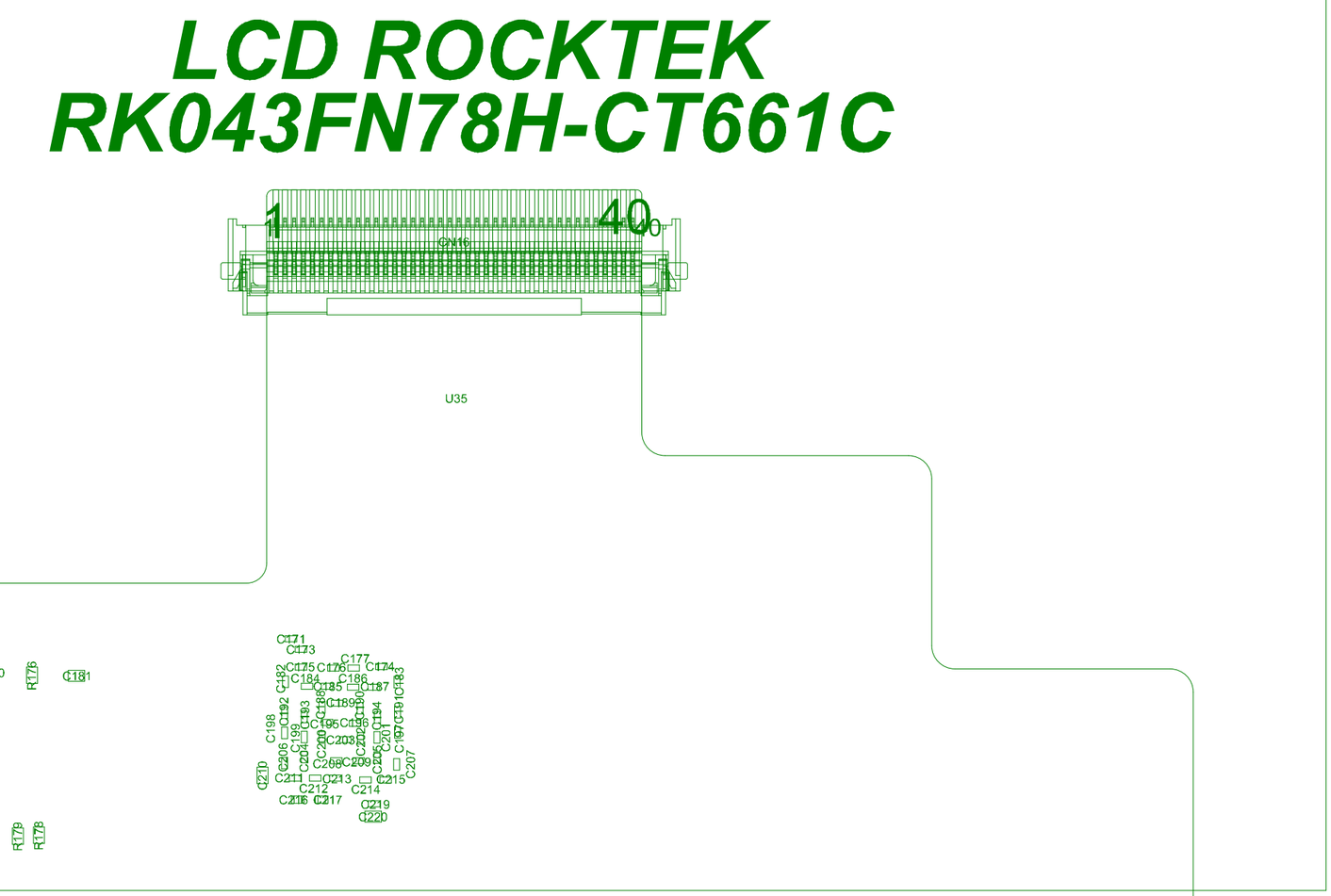
Project: STM32MP135-DK	
Layer: M14-Top Assembly	Gerber: .GM14
Variant: MP135F	Ref: MB1635
Date: 22-JULY-28	Rev: E

Board 25x9 sticker

HW2

LCD ROCKTEK
RK043FN78H-CT661C

U35



CN8

CN10

CN11

CN14

SW1

TR1

TR2

LD1

LD2

LD3

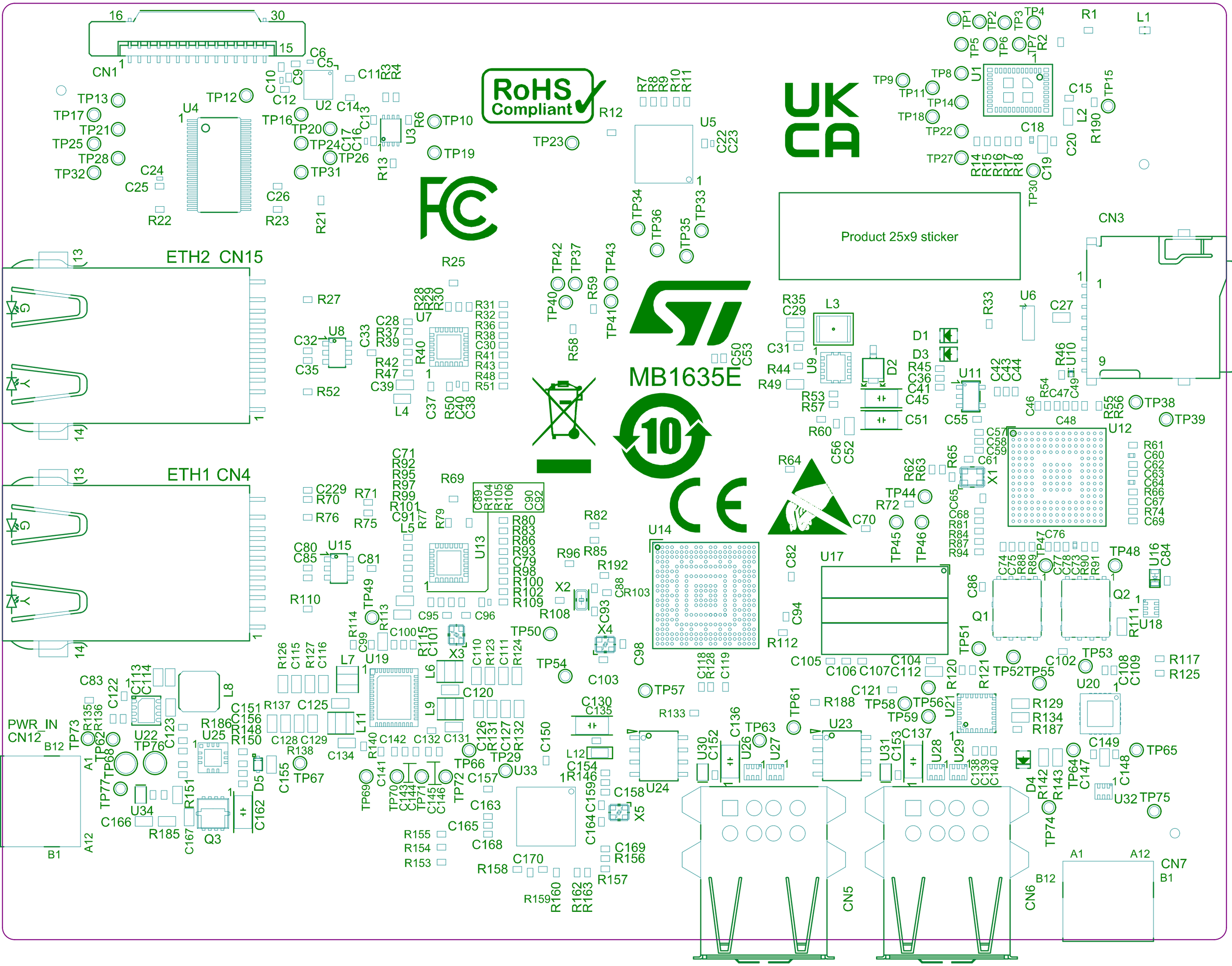
LD4

LD5

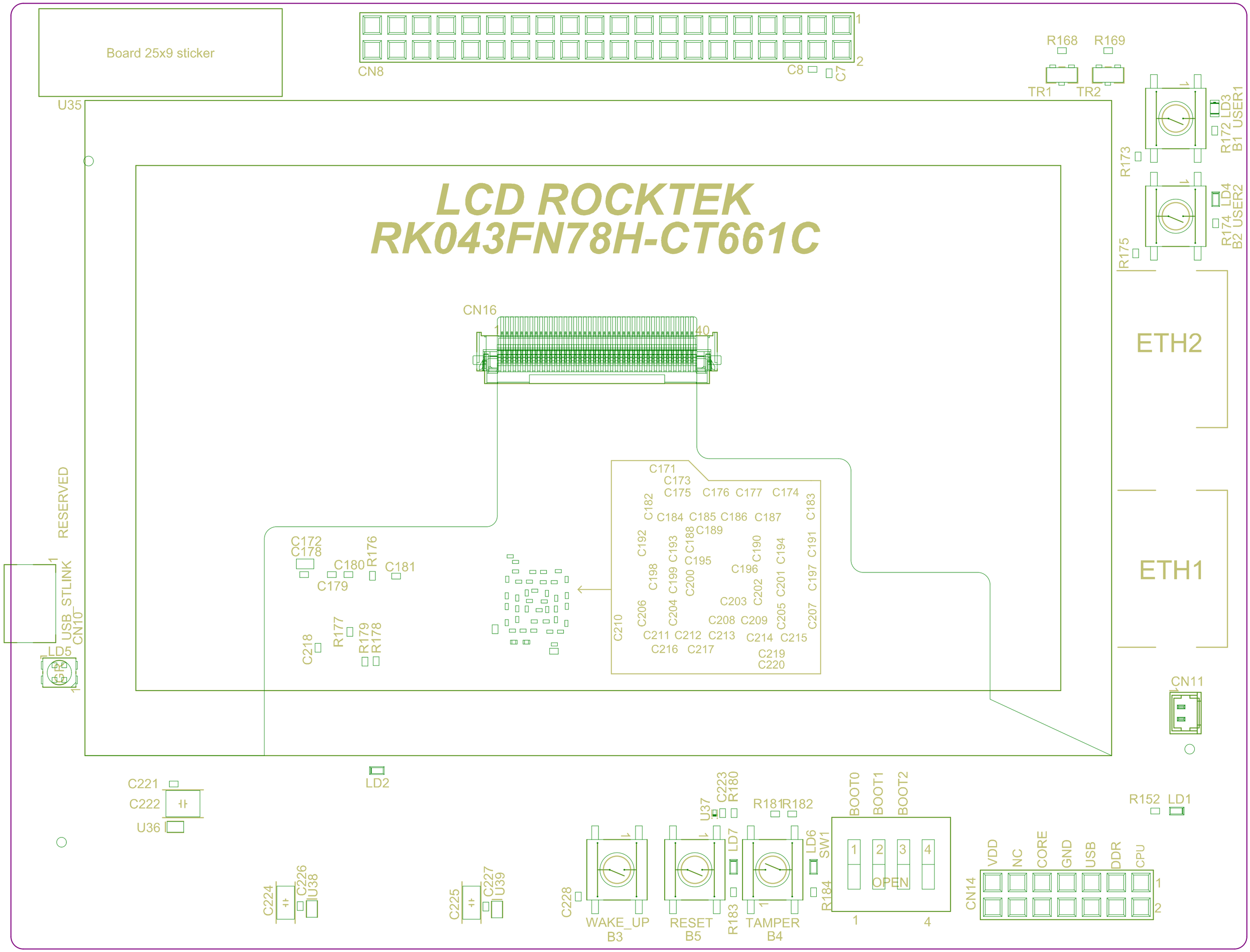
LD6

LD7

Project: STM32MP135-DK	
Layer: M15-Bottom Assembly	Gerber:.GM15
Variant: MP135F	Ref: MB1635
Date: 22-JULY-28	Rev: E



M14-Top Assembly .GM14



M15-Bottom Assembly .GM15